

SHAP-Based Explainable AI Model for Early Detection of Cardiovascular Disease

Md. Abu Rayhan Imran

Department of CSE

Green University of Bangladesh

Narayanganj-1461, Dhaka, Bangladesh

aburayhan2550@gmail.com

Tawhid Hasan

Department of CSE

Green University of Bangladesh

Narayanganj-1461, Dhaka, Bangladesh

tawhidhasan2252@gmail.com

Nahida Akter

Department of CSE

Green University of Bangladesh

Narayanganj-1461, Dhaka, Bangladesh

akternahida625@gmail.com

Md. Solaiman Mia

Department of CSE

Green University of Bangladesh

Narayanganj-1461, Dhaka, Bangladesh

solaiman@cse.green.edu.bd

Tanpia Tasnim

Department of CSE

Green University of Bangladesh

Narayanganj-1461, Dhaka, Bangladesh

tanpia@cse.green.edu.bd

Abstract—Cardiovascular diseases (CVDs) continue to rank among the leading contributors to global mortality. Although significant progress has been made, existing CVD prediction models still face many barriers including imbalanced datasets, limited data sizes, inefficient feature selection and the inability to provide fully effective solutions. This paper introduces an ensemble machine learning framework for the early prediction of CVD, using a publicly available heart disease dataset containing 4,45,132 samples. Feature selection was performed using hybrid Mutual Information scores and Random Forest (RF) importance rankings, and the class imbalance problem was addressed using the Synthetic Minority Oversampling Technique (SMOTE). Our proposed stacking ensemble classifier integrates XGBoost, LightGBM and CatBoost as base learners, with a RF meta-learner combining their outputs. Traditional models including AdaBoost, Gradient Boosting, Logistic Regression, and Linear Support Vector Classifier, were also trained on the same dataset for the comparison. The experimental evaluation shows that our proposed model has achieved an accuracy of 96.6% and an area under the curve of approximately 0.99, which significantly outperforming individual classifiers. To improve interpretability, SHAP (SHapley Additive exPlanations) values were employed, offering clear insights into how features influenced the model. These results demonstrate that the stacking ensemble framework is a robust, explainable and generalizable tool for CVD prediction.

Index Terms—Cardiovascular Disease, SHapley Additive exPlanations (SHAP), Synthetic Minority Oversampling Technique (SMOTE).

I. INTRODUCTION

Cardiovascular diseases (CVDs) represent a broad category of disorders that impair the structure or function of the heart and vascular system. Examples include coronary and cerebrovascular diseases, peripheral arterial disorders, rheumatic and congenital cardiac abnormalities, and blood-clot related conditions such as deep vein thrombosis and pulmonary embolism [1]. According to the World Health Organization

(WHO), CVDs are the leading cause of death worldwide, accounting for nearly 19.8 million deaths in 2022, representing approximately one third of all global deaths [1]. In the United States alone, nearly 8,05,000 cases of heart attack are reported each year, while developing countries face even greater challenges due to limited access to timely diagnosis and treatment [2]. Therefore, early detection of CVD is essential to improve patient prognosis, facilitate preventive measures, and reduce healthcare costs. However, traditional diagnostic processes are heavily based on conventional assessment of clinical and demographic risk factors, which may require considerable time, be inconsistent, and be prone to differences between evaluators [3]. These realities highlight the urgency of improving early detection systems not only to improve patient outcomes but also to reduce healthcare costs.

There has been growing interest in computational healthcare, much of the existing works on heart disease prediction have been carried out using relatively small datasets. These studies provided useful information, their limited data size limits generalization and clinical applicability [4]. In addition, most existing approaches focus solely on predictive accuracy, with little attention to interpretability. Furthermore, much of the prior works focused on individual classifiers and did not incorporate ensemble-based approaches, which limits the ability to capture the complementary strengths of multiple models [5]. Such models often report high accuracy on individual datasets but demonstrate overfitting, poor cross-dataset performance and reduced interpretability, limiting their clinical values.

Not all studies incorporate behavioral predictors, such as physical activity, sleep or vaccination history, which are now recognized as critical determinants of cardiovascular health [6]. As a result, these models rely heavily based on conventional risk indicators such as cholesterol and blood pressure

while ignoring holistic health indicators, thereby reducing predictive robustness. To address the identified limitations in prior studies, this paper makes the following key contributions:

- Proposing a stacking ensemble (SE) classifier integrating Extreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), and Light Gradient Boosting Machine (LightGBM) as base learners, with Random Forest (RF) as the meta-learner, to enhance accuracy and generalization beyond single classifiers.
- Benchmarking multiple algorithms and selecting XGBoost, LightGBM, and CatBoost as the most stable and complementary base learners for a more robust and interpretable stacking model.
- Addressing class imbalance using SMOTE, ensuring balanced learning and improved detection of minority cases.
- Implementing a feature selection strategy using hybrid Mutual Information (MI) and RF-based importance ranking information-theoretic scores, ensuring the retention of the relevant predictors and improving model robustness.
- Introducing an enhanced interpretability with SHAP, providing transparent insights into feature contributions for clinical trust.

II. LITERATURE REVIEW

Recent studies have increasingly explored machine learning (ML) techniques for CVD prediction, recognizing their potential to enhance early diagnosis and prevention.

Singh and Gupta [7] evaluated the performance of several classical ML algorithms, including Logistic Regression (LR), RF, K -Nearest Neighbors (KNN), Naïve Bayes (NB) and Support Vector Machine (SVM), on original, scaled and re-sampled datasets. They reported an accuracy of up to 85% and highlighted the significant impact of preprocessing on model performance, highlighting the critical role of adequate data preprocessing for attaining reliable CVD prediction. But the study did not examine how these methods affect downstream decision reliability or model transparency in a clinical context.

Alfadli and Almagrabi [8] utilized the UCI Heart Disease Dataset, which contains 920 patient cases with 76 attributes (13 commonly used), to predict cardiovascular disease using classical and deep learning models. Their feature-limited hybrid approach achieved 84.24% of accuracy, 89.22% of recall, 83.49% of precision and F1-score around 86%, demonstrating effective early diagnosis with minimal patient data. But the model was trained on a limited dataset without any formal feature selection or interpretability framework, which may hinder clinical adoption.

Asadi et al. [9] applied tree-based models with clustering (GLMM Tree, GMERF) on 9,499 samples with 38 variables, where the GMERF attained optimal results with an area under the curve (AUC) score of 0.80. The most influential predictors were identified as age, low-density lipoprotein (LDL), family history of cardiac disease, physical activity and hypertension. Yet the framework did not include explainable AI tools or account for behavioral and lifestyle variables that could enhance prediction robustness.

In [10], the authors used ensemble learning with SHAP explainability on two public datasets and achieved up to 98% of accuracy showing strong performance and interpretability. However, reliance on limited datasets and retention of highly collinear features raise concerns about generalizability. The study also did not assess the clinical implementation in the real-world or the reliability of the decision beyond the experimental settings.

The authors of [11] applied a weighted KNN with feature grouping and Relief-based weighting on five UCI/Kaggle heart datasets, achieving an accuracy of 93.28%. But they did not validate the grouping logic against clinical taxonomies or provide insights how feature weights impact interpretability across patient types.

Chi et al. [12] introduced the GBDT+LR model for CVD prediction using the UCI CVD dataset (68,283 records, 11 features). The GBDT+LR model combines gradient boosting decision trees (GBDT) for feature extraction with LR for prediction, achieving superior performance in accuracy of 78.3% compared to single models like LR, GBDT, RF and SVM. The study also developed a Spark-based cardiovascular disease prediction platform, although the layered architecture obscures the influence of individual features and omits critical lifestyle predictors that could enhance real-world predictive reliability.

In contrast to individual model-based studies, recent research highlights that the ensemble and hybrid approaches substantially improve cardiovascular disease prediction. Classical methods provide a baseline performance, but techniques such as optimized XGBoost, GMERF, weighted KNN and GBDT+LR consistently achieve higher accuracy, recall, and AUC across diverse datasets. Common predictors including age, LDL, family history, physical activity and hypertension are repeatedly emphasized. These findings collectively suggest that hybrid ensemble frameworks, combined with appropriate preprocessing, represent the effective direction for enhancing reliable early diagnosis of CVD.

III. PROPOSED METHODOLOGY

Our proposed Stacking Ensemble (SE) framework predicts CVD from clinical and demographic data while ensuring reliable and interpretable results through its ensemble architecture. The dataset contains diverse clinical and demographic attributes, which were preprocessed through a comprehensive pipeline including missing value handling, categorical encoding, outlier detection, normalization, hybrid feature selection, hybrid MI and RF, and class balancing. The processed data were divided into training and testing sets with an 80:20 ratio. The SE was then developed using XGBoost, LightGBM and CatBoost as base learners and RF as meta-learner. The performance of our proposed model on the test set was evaluated using standard metrics, and Explainable AI (XAI) techniques were applied to interpret feature contributions. The results confirmed the effectiveness of our proposed framework in predicting cardiovascular disease.

A. Data Collection

This research utilized the *Indicators of Heart Disease (IHD) 2022 Update* dataset from Kaggle, derived from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) [13]. The dataset comprises over 4,45,132 samples and 40 selected features, including demographics and major clinical determinants.

B. Data Preprocessing

The dataset underwent a comprehensive preprocessing pipeline to ensure quality and consistency for predictive modeling. Missing values in numeric features were imputed using the median (or mean for symmetric distributions), while categorical features, such as Yes/No indicators and race, were imputed using the mode or labeled as "Unknown" when necessary. Binary features were encoded as 0/1, ordered categorical features were mapped to integer values, and nominal categories without natural order were transformed using one-hot encoding. Outliers were detected using the Interquartile Range (IQR) and z-score methods, with only obvious errors capped or removed based on percentile thresholds. Finally, continuous features were standardized using the Standard Scaler to ensure uniform scaling. This preprocessing pipeline produced a clean, consistent and representative dataset, forming a robust foundation for accurate cardiovascular disease prediction.

C. Feature Selection

Feature selection represents an essential stage in predictive modeling, as it reduces dimensionality, eliminates irrelevant or redundant variables, and enhances both the computational efficiency and the explanatory capacity of ML models [14]. In this paper, two complementary feature selection methods are applied, named hybrid MI and RF features.

1) *Mutual Information (MI)*: MI measures the dependency across a pair of variables and captures how much information about one variable can be inferred from the other. It is well-suited for capturing both linear and non-linear associations between features and the target variable.

2) *Random Forest Feature Importance (RF-FI)*: RF assigns variable importance scores determined by the reduction in impurity measures (such as Gini index or entropy) across the nodes of DTs tree splits. Features that contribute more to reduce classification error are ranked higher. This method is particularly effective because it can handle both categorical and numerical variables and naturally incorporates feature interactions. Hybrid of MI and RF-FI ensures that both dependency-based and ensemble-based importance measures were captured, leading to a robust selection of predictors as shown in Fig. 1.

When the hybrid one is used together, it becomes easier to filter out the features that add almost no value. Our selected features, including HadAngina, AgeCategory, and GeneralHealth, align well with widely recognized CVD risk factors. These are already known to be strong CVD indicators, which suggests that the combined method kept the meaningful features while cutting down the unnecessary ones.

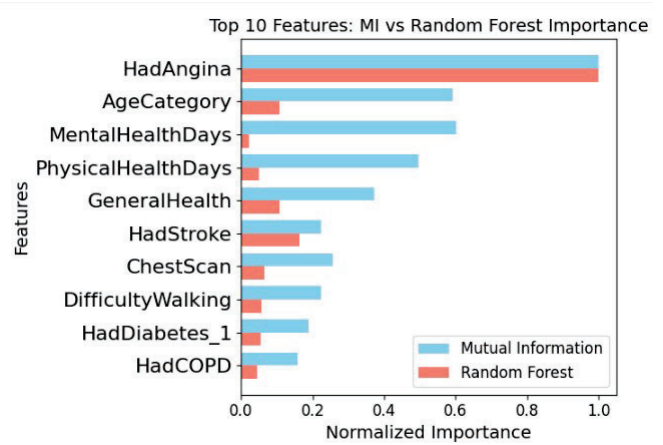


Fig. 1: Feature Importance Ranking Based on MI and RF Methods (Top 10 Features)

D. Handling Class Imbalance with SMOTE

Class imbalance is a common challenge in clinical datasets. In our dataset, there were about 4,01,931 negative cases (no prior heart attack) and only 22,655 positive cases, showing a severe imbalance. Training models on such skewed data leads to bias predictions favoring the majority class. To overcome this issue, we applied the SMOTE, which generates new synthetic minority samples by interpolating between existing ones. Unlike simple oversampling, SMOTE prevents overfitting caused by duplicate records and produces more diverse, realistic samples. This makes it especially suitable for healthcare data, where minority cases (patients with disease) carry higher clinical importance. After applying SMOTE, both classes were balanced to 4,20,024 samples each, ensuring equal representation. Fig. 2 shows that the balancing procedure not only reduced bias toward the majority class but also improved fairness, sensitivity to rare events, and the overall generalization capability of our predictive framework. Despite its benefits, SMOTE has some limitations also. Its synthetic samples generated through interpolation can add noise or unrealistic combinations of features, particularly with categorical or skewed variables, potentially oversmoothing class boundaries and missing real clinical variability in CVD cases.

E. Classification Models

This research employed several state-of-the-art classification algorithms to predict CVD outcomes, including XGBoost, CatBoost, LightGBM, and RF.

1) *Extreme Gradient Boosting (XGBoost)*: XGBoost is a tree-based ensemble method that builds models sequentially [15]. Each new tree is trained to minimize the errors (residuals) of the previous trees using gradient descent on a specified loss function. It also includes regularization terms to reduce overfitting and supports parallel processing for faster computation. By combining multiple weak learners, XGBoost achieves high predictive performance and can capture complex feature interactions [15].

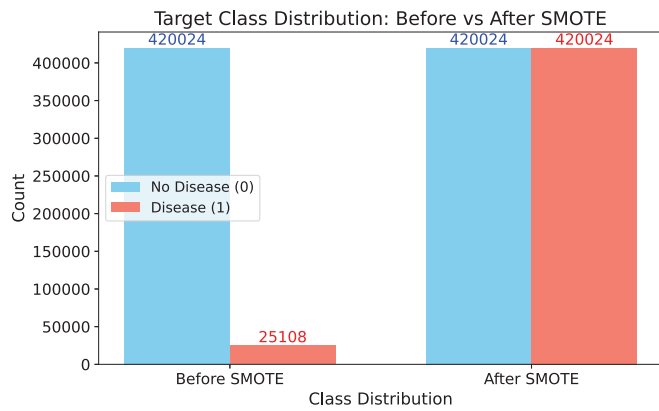


Fig. 2: Distribution of the Target Feature after Balancing with SMOTE

2) *Categorical Boosting (CatBoost)*: CatBoost is a gradient boosting algorithm designed to handle categorical features efficiently [15]. It uses a technique called ordered boosting to prevent target leakage and to reduce overfitting. Categorical features are converted into numerical representations using statistics derived from the training data, eliminating the need for extensive preprocessing. Trees are grown sequentially, optimizing a loss function to minimize prediction errors.

3) *Light Gradient Boosting Machine (LightGBM)*: LightGBM is a highly efficient gradient boosting framework that grows trees leaf-wise instead of level-wise, allowing for deeper splits where they are most needed [7]. It uses histogram-based techniques to reduce memory usage and to speed up training. Like other boosting algorithms, LightGBM adds trees sequentially to correct the errors of previous trees, achieving high accuracy with fast computation.

4) *Random Forest (RF)*: RF is an ensemble learning method that constructs a large number of DTs using bootstrap aggregation [7]. Each tree is trained on a random subset of the data and a random subset of features at each split, reducing variance and preventing overfitting. The final prediction is obtained by averaging (for regression) or majority voting (for classification) across all trees, improving generalization.

F. Proposed Stacking Ensemble (SE) Model

SE is an ensemble learning technique where multiple base learners are trained to make predictions, and their outputs are combined using a meta-learner to produce a final prediction, leveraging the strengths of different models to improve accuracy and reduce errors. In our implementation, the base learners include XGBoost, CatBoost and LightGBM, whose predictions are fed into a RF meta-learner to generate a more accurate final prediction.

To prevent overfitting and to ensure unbiased learning, base learner outputs were generated using K-fold cross-validation, where each model was trained on $K = 5$ folds and evaluated on the hold-out fold. The predictions from these held-out partitions were subsequently used as inputs for training the

meta-learner. Initially, multiple base learners (Level 0) are trained on the dataset to generate predictions, which serve as input features for the Level 1 meta-classifier. The meta-classifier then learns to optimally combine these base predictions to produce the final ensemble output, improving overall predictive performance and reducing the risk of overfitting.

G. Explainable AI (XAI) Using SHAP

To enhance the transparency, interpretability and trustworthiness of the predictive framework, XAI techniques were integrated into the modeling pipeline. Specifically, SHAP were employed to interpret the contribution of individual features to the model's predictions. SHAP provides a game-theoretic framework for feature attribution, offering global and class-level interpretability. Using the top 20 features, it ranks the most influential predictors driving cardiovascular disease outcomes. The class-level interpretation further distinguishes class 0 (disease absent) and class 1 (disease present), improving model transparency and clinical relevance.

H. Inference System

During inference, a new patient record first passes through the same preprocessing steps used during training, including encoding and scaling. The processed input is then forwarded to the three base learners (XGBoost, LightGBM, and CatBoost), each producing an independent probability of cardiovascular disease. These outputs are combined by the RF meta-learner, which generates the final CVD prediction. SHAP values are subsequently computed to show how individual features contributed to the model's decision, allowing clinicians to understand the reasoning behind each prediction.

IV. RESULT ANALYSIS

The performance of the classification models was evaluated using standard metrics, including accuracy, precision, recall, F1-score and AUC. Table I presents a comparative analysis of traditional models such as AdaBoost, GB, LR, LSVC and SGD, with our proposed SE model. Our proposed model outperforms traditional classifiers in terms of accuracy, precision, recall, F1-score, Mean Absolute Error (MAE), and AUC, demonstrating its superior predictive capability. Fig. 3 shows the confusion matrix which indicates robust classification accuracy. The model correctly identified 79,714 True Positive and 82,653 True Negative, while misclassify only 4,291 False Positive and 1,352 False Negative, demonstrating its effectiveness in distinguishing patients with and without cardiovascular disease. Fig. 4 shows the ROC curves which visualize the trade-off between these metrics across various thresholds and exhibit superior discriminative ability compared to the traditional classifiers, with a high AUC value highlighting its excellent performance in predicting cardiovascular disease. Overall, the combined analysis of these evaluation metrics confirms the robustness, reliability and effectiveness of our proposed model for accurate CVD risk prediction.

In Fig. 5, SHAP-based visual interpretation of the stacking ensemble model offers transparent insights into feature influence through combined global and class-level explanations.

TABLE I: Comparison of the Proposed Stacking Ensemble and Traditional Models

Model	Accuracy	Precision	Recall	F1-Score	MAE	AUC
AdaBoost	0.8314	0.830	0.833	0.831	0.1686	0.913
Gradient Boosting	0.8980	0.892	0.904	0.898	0.1020	0.963
Logistic Regression	0.9273	0.947	0.904	0.925	0.0727	0.977
LSVC	0.9260	0.952	0.896	0.923	0.0740	0.980
SGD	0.9265	0.946	0.904	0.924	0.0735	0.976
Proposed SE Model	0.9664	0.983	0.948	0.965	0.0336	0.991

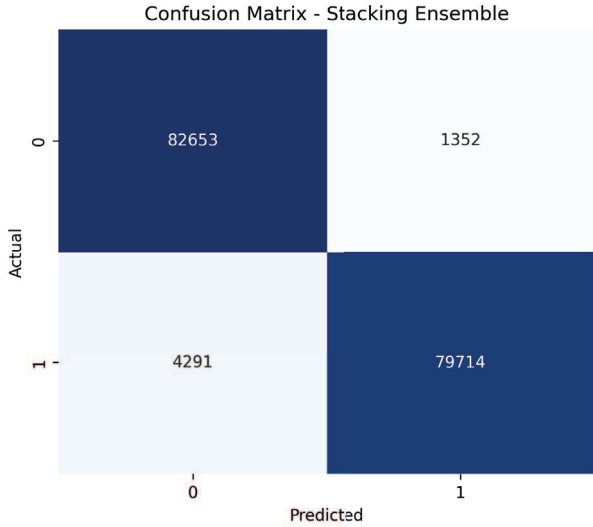


Fig. 3: Confusion Matrix of the Proposed Stacking Ensemble Model

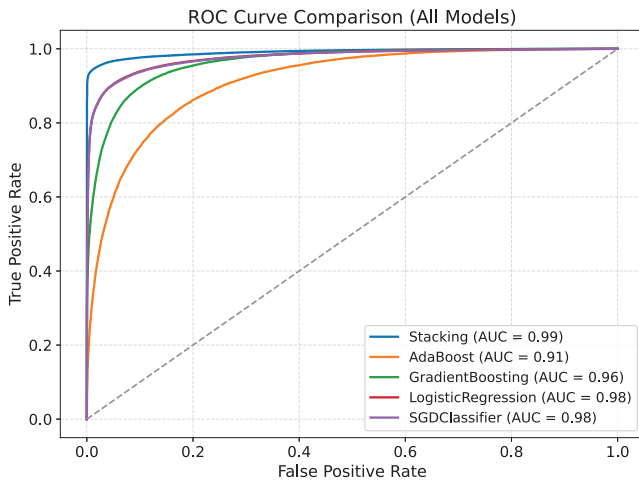


Fig. 4: ROC Curves of the Proposed Stacking Ensemble Model Compared to Traditional Classifiers

The summary dot and bar plots reveal that *HeightInMeters*, *SleepHours*, and *PhysicalHealthDays* exert the highest positive impact on prediction outcomes, whereas features such as *HadAngina* and *LastCheckupTime* exhibit relatively lower contributions. Class-wise distributions (Level 0 and Level 1) further indicate that *HeightInMeters*, *SleepHours*, and *HadAngina* serve as dominant discriminative factors, while *SmokerStatus* and *MentalHealthDays* demonstrate comparatively weak

influence. Overall, the SHAP explanations not only quantify marginal feature contributions but also strengthen model interpretability, increasing clinical reliability and facilitating more informed, evidence-driven decision-making.

Table II shows the comparison of our proposed SE model with the existing works in the literature. Barry et al. [11] reported 98% of accuracy on a relatively small dataset of 1,025 samples. In contrast, our proposed model achieved 96.66% of accuracy on a substantially larger IHD dataset with 4,45,312 samples, which reflects stronger generalization capability and is considered a more reliable and meaningful performance outcome due to the significantly larger and more diverse dataset.

TABLE II: Comparison with the Existing Works

Existing Works	Dataset	Samples	Used Models	Accuracy
[7]	HDD	4,240	RF/SVM/LR	0.85
[8]	UCI Heart Disease	920	Hybrid Ensemble	0.84
[9]	Fasa Cohort (FACS)	9,499	GMERF	0.75
[10]	HDCC	1,190	Stacking Ensemble	0.91
[11]	UHDD	1,025	Stacking Ensemble	0.98
[12]	UCI CVD	68,283	GBDT+LR	0.78
Proposed Work	IHD	4,45,132	Stacking Ensemble	0.96

V. CONCLUSION

CVDs remain a major global health challenge, making early detection critical for reducing mortality and healthcare burdens. This paper presents an ML-based predictive framework using the IHD dataset, comprising over 4,45,132 adult health records with both clinical and lifestyle variables [13]. Experimental evaluation shows that our proposed model achieves an accuracy of 96.6%, precision of 98.3%, recall of 94.8%, F1-score of 96.5%, MAE of 0.034, and AUC of 0.991, outperforming traditional classifiers. These results demonstrate that SEs can serve as a reliable and generalizable tool for CVD prediction by using large-scale data and heterogeneous learners, effectively addressing the limitations of previous studies that relied on small datasets or single classifiers. Despite the strong performance of the proposed framework, several limitations exist. The model relies on the IHD 2022 dataset, which is based on self-reported data and may contain biases or inaccuracies. Although SMOTE was used to address class imbalance, synthetic samples may introduce noise or overfitting. Our future work will focus on integrating XAI LIME techniques to enhance interpretability, deploying the framework as a web-based platform for healthcare professionals, and developing long-term patient monitoring systems for continuous cardiovascular risk assessment. These advances aim to strengthen personalized healthcare and promote the seamless incorporation of ML models into practical clinical decision support systems.

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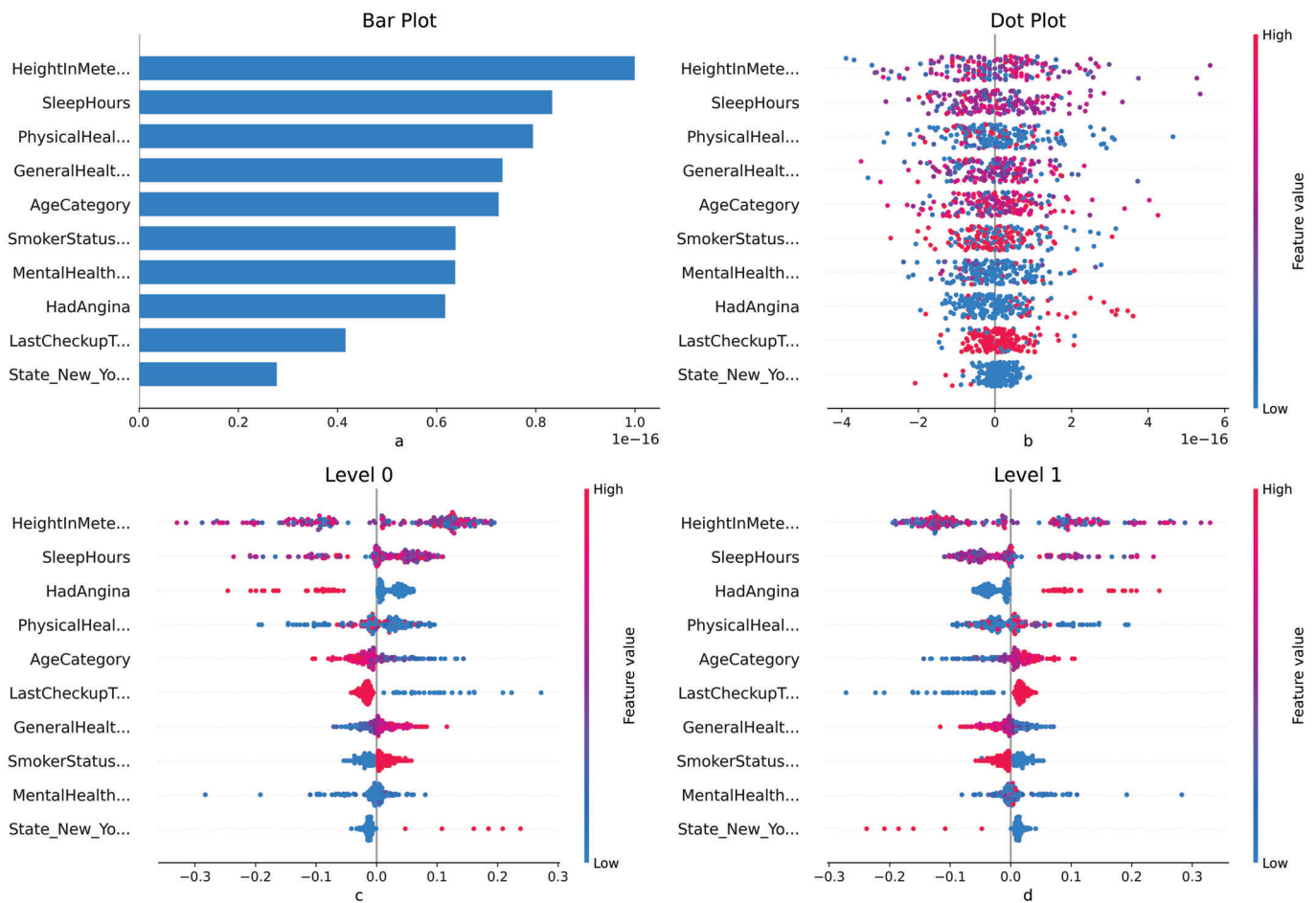


Fig. 5: Interpretable Stacking Ensemble Model Explained by SHAP Values

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